

JHARSUGUDA, India

WMO: 428860

Lat: 21.914N

Lon: 84.050E

Elev: 229

StdP: 98.61

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.0	12.2	0.6	4.0	31.6	2.4	4.6	30.3	2.4	24.2	2.1	24.5	0.0	0	0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.8	42.9	22.4	41.7	22.4	40.2	22.7	27.7	32.3	27.2	31.6	27.0	31.3	1.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.9	29.4	26.2	22.3	29.0	26.0	22.0	28.9	90.2	32.5	88.4	32.0	87.0	31.2	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.4	3.0	2.5	DB	8.4	44.8	1.1	0.9	7.6	45.4	7.0	45.9	6.4	46.4	5.7	47.0	(4)
(5)			WB	7.0	29.7	1.0	1.0	6.2	30.4	5.6	31.0	5.0	31.6	4.3	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	20.3	23.7	28.1	32.4	34.0	31.7	28.2	28.1	28.3	26.9	23.5	20.5	(6)	
	DBStd	4.72	1.95	2.38	2.42	2.39	2.10	3.18	1.49	1.22	1.15	1.63	1.86	1.87	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	8	5	0	0	0	0	0	0	0	0	0	0	3	(9)	
	CDD10.0	6262	319	383	562	672	744	651	564	562	549	524	405	326	(10)	
	CDD18.3	3229	66	150	303	422	486	401	306	304	299	266	155	71	(11)	
	CDH23.3	40437	622	1566	3883	6563	7859	5730	3362	3261	3228	2562	1222	578	(12)	
	CDH26.7	21037	125	697	2239	4352	5422	3433	1156	1034	1133	968	379	99	(13)	
(14) Wind	WSAvg	0.6	0.3	0.4	0.5	0.7	0.9	1.0	0.8	0.8	0.6	0.3	0.2	0.2	(14)	
(15) Precipitation	PrecAvg	1539	14	17	14	23	38	215	483	392	258	68	18	10	(15)	
	PrecMax	2423	101	51	53	92	182	530	908	654	665	214	93	107	(16)	
	PrecMin	918	0	6	2	0	0	73	264	200	112	8	3	0	(17)	
	PrecStd	293	23	11	13	22	35	101	149	125	119	53	23	23	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	35.7	40.1	43.6	44.5	43.6	34.8	33.5	34.0	34.1	31.9	30.0	(19)	
		MCWB	18.1	18.5	19.2	21.2	22.7	23.8	26.9	26.9	26.1	23.9	21.4	18.6	(20)	
	2%	DB	29.3	33.8	38.2	42.2	43.3	42.0	33.5	32.6	33.1	32.9	30.9	28.8	(21)	
		MCWB	17.4	18.5	19.7	20.9	22.6	24.1	26.8	26.8	26.2	24.1	20.8	18.2	(22)	
	5%	DB	28.0	32.4	37.0	41.1	42.0	39.8	32.3	31.7	32.3	32.0	30.0	27.8	(23)	
		MCWB	16.9	18.1	19.6	21.0	22.9	25.0	26.6	26.7	26.2	24.1	20.4	17.7	(24)	
	10%	DB	26.9	31.0	35.5	39.7	40.6	37.5	31.3	30.8	31.4	31.1	29.0	26.7	(25)	
		MCWB	16.6	18.1	19.4	21.2	23.4	25.4	26.5	26.5	26.2	23.8	19.9	17.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.1	24.0	26.2	27.7	28.3	28.0	27.9	27.9	27.0	24.5	21.5	(27)	
		MCDB	25.5	29.3	33.5	36.2	35.8	34.9	31.9	31.5	31.7	30.9	29.2	27.2	(28)	
	2%	WB	19.7	21.1	23.0	24.9	26.9	27.5	27.3	27.4	27.3	26.4	23.5	20.2	(29)	
		MCDB	24.9	28.1	32.6	34.3	35.7	33.8	31.1	30.8	31.0	30.2	27.7	24.9	(30)	
	5%	WB	18.8	20.2	22.1	24.1	26.3	27.0	27.0	27.1	26.9	25.8	22.7	19.3	(31)	
		MCDB	25.1	27.3	31.1	34.6	35.2	32.9	30.7	30.4	30.6	29.5	27.0	24.6	(32)	
	10%	WB	18.0	19.4	21.3	23.5	25.6	26.6	26.7	26.7	26.6	25.3	21.9	18.4	(33)	
		MCDB	24.6	27.0	30.3	34.3	34.7	32.4	30.0	29.7	30.0	28.8	26.2	24.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	14.5	14.6	14.1	12.8	8.9	5.4	5.2	6.2	9.2	12.5	13.9	(35)	
		MCDBR	15.5	16.1	16.1	15.3	14.4	12.3	7.4	6.5	7.7	10.1	12.9	14.4	(36)	
	5% WB	MCWBR	5.5	4.3	4.1	4.2	4.0	2.9	2.4	2.3	2.5	3.2	4.1	5.0	(37)	
		MCWBR	12.4	12.5	13.3	13.5	12.5	9.0	6.3	5.9	6.6	7.7	9.3	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.469	0.473	0.537	0.556	0.616	0.671	0.592	0.567	0.568	0.541	0.518	0.480	(40)	
		taud	2.016	2.041	1.897	1.895	1.799	1.719	1.905	1.988	1.971	1.987	1.946	2.006	(41)	
	Ebn at Noon		786	816	780	771	721	676	732	753	745	744	733	757	(42)	
		Edh at Noon	159	165	199	202	221	237	197	182	182	172	170	156	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.37	5.22	5.99	6.69	6.66	5.18	3.92	3.99	4.41	4.81	4.44	4.11	(44)	
	RadStd		0.27	0.28	0.27	0.29	0.18	0.52	0.48	0.28	0.38	0.43	0.25	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page